

# Madhur N Patel

Infrastructure & Embedded Systems Engineer

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## SUMMARY

Builds and operates self-managed Linux infrastructure, containerized services, and sensor-fusion systems on Raspberry Pi and ESP32. Looking for infrastructure, embedded, or platform engineering work — ideally somewhere that ships physical products.

## PROJECTS

### NavDashboard — Asset & Operations Tracking Platform

live: [nav.madhur.dev](https://nav.madhur.dev)

- Production operations dashboard: devices, inventory, and personnel tracking, with QR label printing and scan-verified gate-pass workflows for physical stock movement.
- Modular async FastAPI backend behind a per-section permission engine (view / edit / manage per account); AI assistant over live operational data via local Ollama inference.
- React 18, TypeScript, FastAPI, PostgreSQL + PostGIS + pgvector, Redis, MinIO, Nginx, Docker Compose

### GPS-Guided Dual-Device Tracking System

- Two Raspberry Pi-controlled gimbals point toward each other by GPS bearing, then hand off to computer vision for fine LED tracking — real-time 9-axis sensor fusion (GPS, IMU, magnetometer) and gimbal control under PID regulation.
- Python, OpenCV, PySerial, Raspberry Pi 4/5

### ESP32 WiFi Drone with Android Controller

- Custom-built ESP32 quadcopter controlled over WiFi from a self-built Android app with real-time telemetry.
- The same self-built app bridges the GPS dual-device tracker and NavDashboard, connecting field hardware to the operations platform.
- ESP32, Python, Android, WiFi

### Self-Hosted Infrastructure — OCI ARM64 + Cloudflare Tunnel

- Self-managed production environment: six containerized services on a 4-core / 24 GB ARM64 Oracle Cloud instance (Open WebUI, n8n, Portainer, code-server, Ollama, Supabase).
- Zero-trust ingress through Cloudflare Tunnel with no public inbound ports; subdomain routing via tunnel ingress rules; monitoring, backups, and TLS certificate management.
- Docker, Docker Compose, Nginx, Linux (Ubuntu), PostgreSQL

### SpeakInsights — Meeting Intelligence Platform

[github.com/pmadhurn/SpeakinsightsV6](https://github.com/pmadhurn/SpeakinsightsV6)

- Meeting platform that runs fully on-premises — LiveKit WebRTC conferencing for up to 20 participants, WhisperX transcription with speaker attribution, Ollama summarization, and RAG chat over meeting history; no third-party API calls.
- React 18, TypeScript, FastAPI, PostgreSQL + pgvector, Redis, Docker

## EXPERIENCE

### Self-Directed Infrastructure Engineering

2022 – Present

- Implemented CI/CD pipelines with GitHub Actions, automating build, test, and deployment of applications.
- Engineered secure network access with Cloudflare Tunnel, exposing internal services without opening inbound ports to the public internet.

## EDUCATION & CERTIFICATIONS

### B.Tech, Computer Science and Engineering — Indus University, Ahmedabad

2022 – 2026

GPA 8.5/10

AWS Academy: Machine Learning Foundations (Aug 2025) · Cloud Foundations (Jul 2025)

## SKILLS

**Infrastructure:** Docker, Docker Compose, Cloudflare Tunnel, Nginx, Oracle Cloud, Vercel, CI/CD, AWS, GCP

**Embedded/IoT:** Raspberry Pi, ESP32, Arduino, Serial / I2C · **Systems:** Linux (Ubuntu)

**Languages:** Python, JavaScript, TypeScript, SQL, C, C++, Java, Bash

**Backend:** FastAPI, Node.js, Express.js, REST APIs · **Databases:** PostgreSQL, MongoDB, MySQL, SQLite, Redis

**AI/ML:** OpenCV, Ollama (local inference), Pandas, NumPy, TensorFlow, scikit-learn